GIBBS SAMPLING FROM SIGMA-ALGEBRAS

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Abstract. We describe Gibbs sampling intrinsically in terms of conditional expectations onto sub-σ-algebras. In this formulation, covariance under invertible blockwise reparameterisation follows from the behaviour of conditional expectation under measurable changes of variables. The same viewpoint then applies to more general dynamics built from conditional expectations.

1. Gibbs updates associated with sub-σ-algebras

Let $(E, \mathcal{F}, \pi)$ be a probability space, and let $\mathcal{G}_1, \ldots, \mathcal{G}_m \subseteq \mathcal{F}$ be sub-σ-algebras. When a pointwise Markov kernel is invoked below, we assume that E is a standard Borel space and $\mathcal{F}$ is its Borel σ-algebra, so that regular conditional distributions exist. All identities involving conditional expectations are understood almost surely. Each $\mathcal{G}_i$ specifies the information retained by update i; the remaining randomness is resampled according to its conditional distribution under π. For each i, define the operator

$$P_i f := \mathbf{E}_\pi \left[f \mid \mathcal{G}_i \right], \qquad f \in L^1(\pi).$$

Definition 1. (Heat-bath/Gibbs update) We call P_i the Gibbs, or heat-bath, update associated with $\mathcal{G}_i$. Equivalently, a regular conditional distribution K_i is specified (up to versions) by $K_i(x, A) = \mathbf{E}_\pi \left[\mathbf{1}_A \mid \mathcal{G}_i \right](x)$ for $A \in \mathcal{F}$, and

$$P_i f(x) = \int_E f(y)\ K_i(x, \mathrm{d}y).$$

The update preserves precisely the information contained in $\mathcal{G}_i$, and resamples all remaining randomness according to its conditional law under π. Each P_i is a π-invariant Markov operator. On $L^2(\pi)$ it is the orthogonal projection onto $L^2(\mathcal{G}_i, \pi)$, and hence $P_i^2 = P_i$, $P_i^\star = P_i$.

The family $\{P_i\}_{i=1}^m$ is an intrinsic Gibbs system associated with $(\pi, \{\mathcal{G}_i\}_{i=1}^m)$. A particular Gibbs sampler is obtained by specifying a scan rule. For example, if $p_i \geq 0$ and $\sum_{i=1}^m p_i = 1$, then $P_{\mathrm{rand}} = \sum_{i=1}^m p_i P_i$ is the random-scan Gibbs kernel, while $P_{\mathrm{sys}} = P_{i_1} P_{i_2} \cdots P_{i_r}$ is a deterministic scan.

For 'ordinary' block Gibbs sampling on the product space $E = E_1 \times \cdots \times E_m$, with coordinate maps $X_j(x) = x_j$, where block i is updated conditionally on all other blocks, one takes $\mathcal{G}_i = \sigma(X_j : j \neq i)$. Conditioning σ-algebras that agree modulo π-null sets define the same update.

2. Covariance under measurable reparameterisation

Conditional expectation commutes with an invertible change of variables, provided the conditioning σ-algebra is transported along with it. Let $\left(\widetilde{E}, \widetilde{\mathcal{F}}, \widetilde{\pi}\right)$ be a probability space and let $T : E \to \widetilde{E}$ be a bimeasurable bijection such that $\widetilde{\pi} = T_\# \pi$. Define the (unitary) pullback operator

$$U_T : L^2(\widetilde{\pi}) \to L^2(\pi), \qquad U_T g = g \circ T.$$

Suppose that $\widetilde{\mathcal{G}}_i \subseteq \widetilde{\mathcal{F}}$ satisfies $T^{-1}\widetilde{\mathcal{G}}_i = \mathcal{G}_i$, and write $\widetilde{P}_i = \mathbf{E}_{\widetilde{\pi}}\left[\cdot \mid \widetilde{\mathcal{G}}_i \right]$.

Proposition 2. *(Covariance of Gibbs updates) For every i,*

$$U_T \widetilde{P}_i = P_i U_T,$$

or equivalently,

$$\widetilde{P}_i = U_T^{-1} P_i U_T.$$

Proof. Let $g \in L^2(\widetilde{\pi})$. Since $\widetilde{P}_i g$ is $\widetilde{\mathcal{G}}_i$-measurable, $\left(\widetilde{P}_i g\right) \circ T$ is $\mathcal{G}_i$-measurable. Moreover, for any $A \in \mathcal{G}_i$, writing $A = T^{-1}\widetilde{A}$ with $\widetilde{A} \in \widetilde{\mathcal{G}}_i$,

$$\int_A \left(\widetilde{P}_i g\right) \circ T \, \mathrm{d}\pi = \int_{\widetilde{A}} \widetilde{P}_i g \, \mathrm{d}\widetilde{\pi} = \int_{\widetilde{A}} g \, \mathrm{d}\widetilde{\pi} = \int_A g \circ T \, \mathrm{d}\pi.$$

By the defining property of conditional expectation,

$$\left(\widetilde{P}_i g\right) \circ T = \mathbf{E}_\pi \left[g \circ T \mid \mathcal{G}_i\right] = P_i \left(U_T g\right),$$

and hence we conclude that $U_T \widetilde{P}_i = P_i U_T$. □

For the transition kernels, this gives $\widetilde{K}_i(Tx, \cdot) = T_{\#} K_i(x, \cdot)$ for π-almost every x. To obtain a correspondence for every initial distribution, it suffices to choose the transformed versions by

$$\widetilde{K}_i(y, B) := K_i\left(T^{-1}y, T^{-1}B\right), \qquad B \in \widetilde{\mathcal{F}}.$$

With these versions and the same scan rule, $T(X_n)$ is the transformed Gibbs chain whenever (X_n) is the original chain.

3. Blockwise-invertible reparameterisation

Consider now the product space $E = E_1 \times \cdots \times E_m$ equipped with the product σ-algebra and coordinate maps $X_j(x) = x_j$ and the ordinary block Gibbs fields $\mathcal{G}_i = \sigma(X_j : j \neq i)$. For each block let $T_j : E_j \to \widetilde{E}_j$ be a bimeasurable bijection, put $\widetilde{E} = \widetilde{E}_1 \times \cdots \times \widetilde{E}_m$, and define

$$T(x_1, \ldots, x_m) = \left(T_1(x_1), \ldots, T_m(x_m)\right).$$

Equip the target product with its product σ-algebra. Let $\widetilde{\pi} = T_{\#}\pi$, and let $\widetilde{X}_j(y) = y_j$ denote the coordinate maps on $\widetilde{E}$. Define $\widetilde{\mathcal{G}}_i = \sigma\left(\widetilde{X}_j : j \neq i\right)$. Since each T_j is invertible,

$$T^{-1}\widetilde{\mathcal{G}}_i = \sigma\left(\widetilde{X}_j \circ T : j \neq i\right) = \sigma\left(T_j \circ X_j : j \neq i\right) = \mathcal{G}_i.$$

The proposition therefore gives $\widetilde{P}_i = U_T^{-1} P_i U_T$, so exact block Gibbs sampling is *covariant under arbitrary invertible reparameterisations performed separately within the blocks.*

In particular, if $P = P_{i_1} \cdots P_{i_r}$, then $\widetilde{P} = U_T^{-1} P U_T$, and similarly, if $P = \sum_i p_i P_i$, then $\widetilde{P} = U_T^{-1} P U_T$. Hence corresponding chains have identical probabilistic behaviour up to the deterministic change of variables T. In particular, all unitarily invariant L^2 quantities, such as the spectrum, operator norms, and (in reversible cases) the spectral gap, are unchanged. The stationary autocorrelation structure is likewise identical for corresponding observables.

Remark 3. The coordinatewise assumption on the transformation T is not incidental. For a general invertible transformation mixing several blocks, $T^{-1}\sigma\left(\widetilde{X}_j : j \neq i\right)$ need not equal $\sigma(X_j : j \neq i)$. The pushed-forward chain is still conjugate to the original Markov chain, but it need no longer be the ordinary coordinate Gibbs sampler for $T_{\#}\pi$. It is instead a Gibbs sampler relative to the transported family of sub-σ-algebras.

Remark 4. The same argument is not specific to coordinate-based Gibbs sampling. Suppose more generally that an update preserves some measurable statistic $S_i : E \to \mathsf{S}_i$ and redraws from the conditional equilibrium distribution given $S_i(X)$. Its Markov operator is $P_i = \mathbf{E}_\pi\left[\cdot \mid \sigma(S_i)\right]$. Given a measurable statistic $\widetilde{S}_i : \widetilde{E} \to \widetilde{\mathsf{S}}_i$, any measurable isomorphism T which transports these information fields, $\sigma\left(\widetilde{S}_i \circ T\right) = \sigma(S_i)$, similarly conjugates the corresponding heat-bath dynamics.

Relation to earlier work

Chang and Pollard [1, Example 13, pp. 311–312] already formulate Gibbs sampling through general conditional resampling. They associate a move with each measurable map admitting a disintegration of the target measure, and form Gibbs samplers by composing moves of this type. They explicitly allow maps other than coordinate projections and state spaces without a product structure.

References

[1] Chang, J. T. and Pollard, D. (1997). Conditioning as disintegration. *Statistica Neerlandica* 51(3), 287–317. DOI: 10.1111/1467-9574.00056.